



Student Number

St. Catherine's School, Waverley

2014

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14, show relevant mathematical reasoning and/or calculations
- Task weighting – 40%

Section I

Pages 3-6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided.

Section II

Pages 7-13

60 marks

- Attempt Questions 11-14
- Allow about 1 hour and 45 minutes for this section
- Answer each question in the booklet provided.

Total Marks – 70

Extension 1 trials.

1 If $y = \sin^{-1}(x^2)$, then $\frac{dy}{dx} =$

A $\frac{2x}{\sqrt{1-x^4}}$

B $\frac{2x}{\sqrt{1-x^2}}$

C $2x \cos^{-1}(x^2)$

D $\frac{2x}{1-x^4}$

2 $\int \sin^2 3x \, dx$ is

A $\cos^2 3x + C$

B $2 \sin 3x \cos 3x + C$

C $\frac{1}{2} \left(x - \frac{\cos 6x}{6} \right) + C$

D $\frac{1}{2} \left(x - \frac{\sin 6x}{6} \right) + C$

3 The primitive of $\sqrt{e^{3x}}$ is given by

A $\frac{3}{2\sqrt{e^{3x}}} + C$

B $\frac{3}{2} e^{\frac{3x}{2}} + C$

C $\frac{2}{3} e^{\frac{3x}{2}} + C$

D $\frac{1}{2} e^{3x} + C$

4 $\int \frac{dx}{\sqrt{1-4x^2}}$ is given by

A $\sin^{-1}2x + C$

B $\sin^{-1}\frac{x}{2} + C$

C $4 \sin^{-1}2x + C$

D $\frac{1}{2} \sin^{-1}2x + C$

5 The equation of motion of a particle moving in Simple harmonic Motion is given by

$\ddot{x} = 1 - 3x$, which of the following statements is true?

A The period of motion is $\frac{2\pi}{3}$ and the centre is $x=\frac{1}{3}$

B The period of motion is $\frac{-2\pi}{3}$ and the centre is $x=3$

C The period of motion is $\frac{2\pi}{3}$ and the centre is $x=3$

D The period of motion is $\frac{2\pi}{\sqrt{3}}$ and the centre is $x=\frac{1}{3}$

6 Given that $\frac{f'(x)}{f(x)} = 1$, which of the following statements is true?

(note: C is a constant in each case)

A $f(x) = \ln x + C$

B $f(x) = e^x + C$

C $f(x) = C e^x$

D $f(x) = C \ln x$

7 Consider the graph of the function $y = \frac{x^2+1}{x}$. The equations of the asymptotes are

A $x = 0$ and $x = 1$

B $x = 0$ and $y = x$

C $x = -1$ and $x = 1$

D No asymptotes

8 The solution to the inequality $x(x^2 - 4) > 0$ is

A $-2 < x < 0$ or $x > 2$

B $-2 < x < 2$

C $x < -2$ or $x > 2$

D $x < -2$ or $0 < x < 2$

9 Surface area S of a spherical balloon is given by the formula $S = 4\pi r^2$,

where r is the radius.

A spherical balloon is being inflated so that, $\frac{dr}{dt} = 2$ cm/sec.

The value of $\frac{dS}{dt}$, when the surface area is $16\pi \text{ cm}^2$ is given by

- A 32π
- B $\frac{16}{\pi}$
- C $256\pi^2$
- D No sufficient information

10 $\cos^{-1}\left(\cos \frac{11\pi}{6}\right)$ is

- A $\frac{11\pi}{6}$
- B $\frac{\sqrt{3}}{2}$
- C $\frac{7\pi}{6}$
- D $\frac{\pi}{6}$

Question 11 Start a new page

- a) Solve for x : $\frac{1}{x-1} \geq 5$ 3
- b) Find the value of k : $x^{3k+4} = e^{8 \ln x}$ 2
- c) Find the ratio in which P divides the interval AB, where
P is $(2, \frac{20}{3})$ A is $(1, 5)$ and B is $(4, 10)$. 3
- d The lines $y = 3mx + 1$ and $y = mx$ are inclined at an angle of α , where
 $\tan \alpha = \frac{1}{2}$.
- (i) Show that $3m^2 - 4m + 1 = 0$ 2
- (ii) Hence find the possible values of m . 1
- e If $x^2 - x - 2$ is a factor of the polynomial $x^4 + 3x^3 + ax^2 - 2x - b$, find the values
of a and b . 4

Question 12 Start a new page

a Show that $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{5} = \tan^{-1}\frac{4}{7}$ 3

b Use the substitution $x = 2 \sin \theta$, to evaluate $\int_0^2 \sqrt{4 - x^2} \, dx$ 4

c Find the general solution to the equation $\sin 2\theta = \cos \theta$ 3

d (i) Show that $\frac{1-x^2}{1+x^2} = -1 + \frac{2}{x^2+1}$ 1

(ii) Hence or otherwise clearly sketch the graph of the function 4

$y = \frac{1-x^2}{1+x^2}$, locating any stationary points and equations of asymptotes.

(Use at least one third of a page)

Question 13 Start a new page

- a Find $\int \frac{5}{1+16x^2} dx$ 2
- b Find the constant term in the expansion of $(2x - \frac{1}{x^2})^9$ 3
- c Consider the expansion of $(3 + 4x)^{12}$ in ascending powers of x.
- (i) Show that $\frac{\text{coefficient of } t_{r+1}}{\text{coefficient of } t_r} = \frac{4(13-r)}{3r}$, where t_r is the r^{th} term. 2
- (ii) Hence or otherwise find the greatest coefficient in the expansion of 2
 $(3 + 4x)^{12}$
- d A particle is projected with velocity 20 metres per second.
This hits a target at a horizontal distance of 20 metres and a vertical height of 10 metres. Take the acceleration due to gravity as 10 m per sec^2
- (i) Show that the equations of motion are given by 2
 $x = 20 \cos \alpha t$ and $y = -5t^2 + 20 \sin \alpha t$, where α , is the angle of projection.
- (ii) Show that the Cartesian equation of the motion is 2

$$y = x \tan \alpha - \frac{x^2}{80} \sec^2 \alpha$$
- (iii) Hence find the values of α 2

Question 14 Start a new page

- a A particle moves with an acceleration given by the expression $a = 7x$. Initially the particle starts from the origin with a velocity of $-3 \text{ metres per second}$. Find an expression for the velocity in terms of the displacement. 3
- b The displacement of a particle (in cm) from a point O on a line after t seconds is given by $x = 3 \sin(2t + \alpha)$. Initially the particle is at $x = 1.5$.
- (i) Find the value of α . 1
 - (ii) Find the acceleration $\ddot{x}$ in terms of the displacement x . 1
 - (iii) Find the time the particle takes to reach the point $x = 0$, for the first time. 2
 - (iii) Find the time it takes to reach an acceleration of -12 cm per sec^2 , for the first time. 2
- c A function is defined as $f(x) = 1 - \cos \frac{x}{2}$.
- (i) State the period of this function. 1
 - (ii) Find the largest domain of the function for which the inverse function $f^{-1}(x)$ exists. Include $x = 0$ in the domain. 1
 - (iii) Find the equation of $y = f^{-1}(x)$. 2
 - (iv) Sketch $y = f^{-1}(x)$. 2

End of Task

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

☐
$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

☐
$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

☐
$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

☐
$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

☐
$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

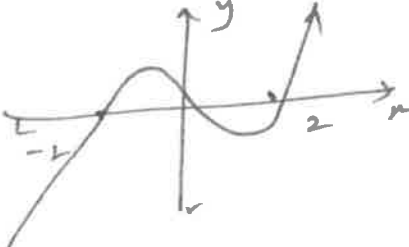
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$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

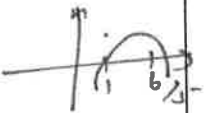
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$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

☐
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

☐
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Qn	Solutions	Marks	Comments: Criteria
	Multiple choice.		
1.	$y = \sin^{-1}(x^2)$ $y' = \frac{(2x)}{\sqrt{1-x^4}}$ A		
2.	$\int \sin^2 3x \, dx = \frac{1}{2} \int (1 - \cos 6x) \, dx$ $= \frac{1}{2} \left(x - \frac{\sin 6x}{6} \right) + C$ D.		
3.	$\int e^{\frac{3x}{2}} \, dx = \frac{2}{3} e^{\frac{3x}{2}} + C.$ (C)		
4.	$\int \frac{dx}{\sqrt{1-4x^2}} = \int \frac{dx}{\sqrt{4(\frac{1}{4}-x^2)}}$ $= \frac{1}{2} \cdot \sin^{-1} 2x + C.$ (D)		
5.	$\ddot{x} = 1 - 3x$ $= -3 \left(x - \frac{1}{3} \right)$ $\frac{2\pi}{\sqrt{3}} \therefore \text{center } x = \frac{1}{3}.$ (D)		
6.	$\int \frac{f'(x)}{f(x)} \, dx = \int 1 \, dx$ $\ln f(x) = x + C.$ $f(x) = e^{x+C}$ $= A e^x$ $\text{or } C e^x$ (C)		

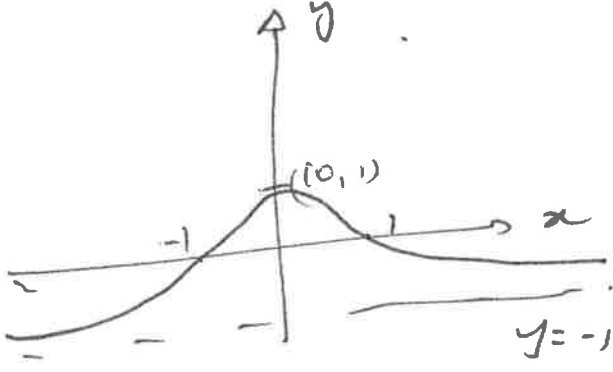
Qn	Solutions	Marks	Comments: Criteria
7	$y = \frac{x^2 + 1}{x}$ $= x + \frac{1}{x} \quad \begin{matrix} x \neq 0 \\ y \neq x \end{matrix}$ (B)		
8	$x(x-2)(x+2) > 0$ $-2 < x < 0 \text{ or } x > 2$  (A)		
9	$S = 4\pi r^2$ $\frac{ds}{dr} = 8\pi r \frac{dr}{dr}$ $= 8 \times \pi \times 2 \times 2$ $= 32\pi$ (A)	$36\pi = 4\pi r^2$ $4 = r^2$ $\underline{2 = r}$	
10	$\cos^{-1}\left(\cos \frac{\pi}{6}\right)$ $= \cos^{-1}\left(\cos\left(2\pi - \frac{\pi}{6}\right)\right)$ $= \cos^{-1}\cos\left(\frac{\pi}{6}\right)$ $= \frac{\pi}{6}$ (D)		

Qn	Solutions	Marks	Comments: Criteria
11.	$\frac{1}{x-1} \geq 5 \quad ; \quad x \neq 1$ $(x-1)^2 \cdot \frac{1}{x-1} \geq 5(x-1)^2$ $x-1 \geq 5(x-1)^2$ $(x-1) - 5(x-1)^2 \geq 0$ $(x-1)(1-5(x-1)) \geq 0$ $(x-1)(6-5x) \geq 0$ $1 < x \leq \frac{6}{5}$ 		$x \neq 1 \quad (-\frac{1}{2})$
b).	$x^{3k+4} = e^{8 \ln x}$ $x^{3k+4} = e^{\ln x^8}$ $x^{3k+4} = x^8$ $3k+4 = 8$ $3k = 4$ $k = \frac{4}{3}$	 1 1	Taking logs. on both sides. ① sol ①
c).	A(1, 5) B(4, 10) k 1, $(2, \frac{20}{3})$ $\frac{4k+1}{k+1} = 2 \quad \left \quad \begin{array}{l} 2k = 1 \\ k = \frac{1}{2} \end{array} \right.$ $4k+1 = 2k+2$		Ensure you take k: 1 rather than k: 1 for each of calculations. A number of students had the wrong formula.

Qn	Solutions	Marks	Comments: Criteria
	<u>Note</u> : works for y. coordinate no need to do this as well. $\frac{10k+5}{k+1} = \frac{20}{3}$ $30k+15 = 20k+20$ $5 = 10k$ $k = \frac{1}{2}$		
d)	$y = 3x^2 + 1$ $y = mx^2$ grad $m_1 = 3m$ grad $m_2 = m$ and $= \frac{3m-m}{1+3m^2}$ $\frac{1}{2} = \frac{3m}{1+3m^2}$ $3m^2 + 1 = 4m$ $3m^2 - 4m + 1 = 0$ $(3m-1)(m-1) = 0$ $m = 1$ $m = \frac{1}{3}$		
e)	$(x-2)(x+1)$ is a factor $\therefore (x-2)$ is a factor of $P(x)$ $P(2) = 0$ $16 + 24 + 4a - 4 - b = 0$ $4a - b = -36$ — (1) $(x+1)$ is a factor : $P(-1) = 0$ $1 - 3 + a + 2 - b = 0$ $a - b = 0$ — (2)		

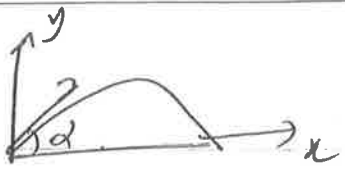
Qn	Solutions	Marks	Comments: Criteria
Q.12	$3a = -36$ $a = -12$ $b = -12$ $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{5} = \tan^{-1} \frac{4}{7}$ $\alpha = \tan^{-1} \frac{1}{3}$ $\tan \alpha = \frac{1}{3}$ $\beta = \tan^{-1} \frac{1}{5}$ $\tan \beta = \frac{1}{5}$ Consider $\tan(\alpha + \beta)$ $= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$ $= \frac{\frac{1}{3} + \frac{1}{5}}{1 - \frac{1}{15}} = \frac{\frac{8}{15}}{\frac{14}{15}} = \frac{4}{7}$ $\therefore \tan^{-1} \frac{4}{7} = \alpha + \beta$ $= \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{5}$ b) $x = 2 \sin \theta$ $dx = 2 \cos \theta d\theta$ $a=0 : \theta = 0$ $\sin \theta = 1$ $x=2 : \theta = \frac{\pi}{2}$ $\therefore \int_0^2 \sqrt{4-x^2} dx = \int_0^{\pi/2} \sqrt{4-4\sin^2 \theta} \cdot 2 \cos \theta d\theta$ $= 4 \int_0^{\pi/2} \cos^2 \theta d\theta$ $= 2 \int_0^{\pi/2} (1 + \cos 2\theta) d\theta$	1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ (2M) $\frac{1}{2}$	

Qn	Solutions	Marks	Comments: Criteria
	$= 2 \left[0 + \frac{\sin 2\theta}{2} \right]^{1/2}$ $= 2 \left[\frac{\pi}{2} \right] = \pi$	$\frac{1}{2}$	-1 if they didn't use $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$
c)	$\sin 2\theta = \cos \theta$ $2 \sin \theta \cos \theta = \cos \theta$ $\cos \theta (2 \sin \theta - 1) = 0$ $\cos \theta = 0 \quad \sin \theta = \frac{1}{2}$ $\theta = 2n\pi \pm \frac{\pi}{2} \quad \theta = n\pi + (-1)^n \cdot \frac{\pi}{6}$	1	losing $\cos \theta$ altogether. $2 \sin \theta \cos \theta = \cos \theta$ $2 \sin \theta = 1$ $\sin \theta = \frac{1}{2}$
d)	$\frac{1-x^2}{1+x^2} = -1 + \frac{2}{x^2+1}$ $-1 + \frac{2}{x^2+1} = \frac{-x^2-1+2}{x^2+1}$ $= \frac{1-x^2}{1+x^2}$ $y = -1 + \frac{2}{x^2+1}$ $y' = \frac{-4x}{(x^2+1)^2}$ At stat pt: $y' = 0$ $x = 0$ $y = 1$ $(0, 1)$ is a stat. pt.		A number of them had wrong information but correct graph. Marks were awarded for the question asked for the graph. If the question was calculated, for the stat. pt, a number of them would have got that point wrong.

Qn	Solutions	Marks	Comments: Criteria
	$\begin{array}{cccc} x & 0^- & 0 & 0^+ \\ y' & > 0 & 0 & < 0 \end{array}$ $\therefore (0, 1) \text{ is a } \text{stat. max. r. pr.}$ $y' = 0 : x = \pm 1$ $x = 0 : y = 1$ $y = -1 + \frac{2}{x^2 + 1}$ $\frac{2}{x^2 + 1} \neq 0 \quad \therefore y = -1 \text{ is an asymptote}$ 		

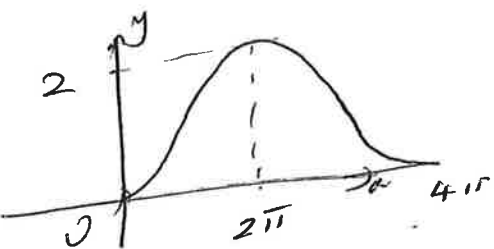
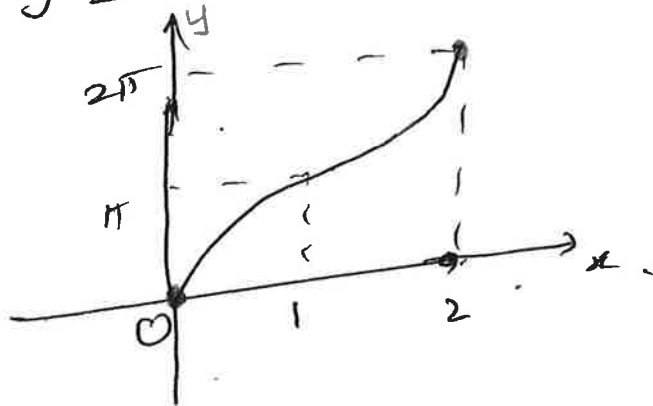
Qn	Solutions	Marks	Comments: Criteria
8.13			
a)	$\int \frac{5}{1+16x^2} dx$ $= \frac{5}{16} \int \frac{1}{\frac{1}{16} + x^2} dx$ $= \frac{5}{16} \left[4 \tan^{-1} 4x + C \right]$ $= \frac{5}{4} \tan^{-1}(4x) + C$	2	1 For $\tan^{-1}(4x) + C$
b)	$(2x - \frac{1}{x^2})^9$ $= {}^9C_0 (2x)^9 + {}^9C_1 (2x)^8 \left(-\frac{1}{x^2}\right) + \dots$ $T_{r+1} = {}^9C_r (2x)^{9-r} \left(-\frac{1}{x^2}\right)^r$ When T_{r+1} is a const: $T_{r+1} = {}^9C_r \cdot 2^{9-r} (-1)^r \cdot x^{9-r-2r}$ T_{r+1} is a const. for $9-3r=0$ $r=3$ $\therefore$ Constant term is $(-1)^3 {}^9C_3 2^6 = -{}^9C_3 \cdot 2^6$ $= -5376$	14	
c)	$(3+4x)^{12} = {}^{12}C_0 3^{12} (4x)^0 + {}^{12}C_1 3^{11} (4x)^1 + \dots$ $T_{r+1} = {}^{12}C_r 3^{12-r} (4x)^r$ $T_r = {}^{12}C_{r-1} 3^{12-(r-1)} (4x)^{r-1}$	14	$-\frac{1}{2}$ for incorrect sign. $\frac{1}{2}$ For correct Expressions for T_{r+1} AND T_r

Qn	Solutions	Marks	Comments: Criteria
	$\frac{\text{Coeff of } t_{r+1}}{\text{Coeff of } t_r} = \frac{{}^{12}C_r}{{}^{12}C_{r-1}} \cdot \frac{3^{12-r}}{3^{13-r}} \cdot \frac{4^r}{4^{r-1}}$ $= \frac{{}^{12}C_r}{{}^{12}C_{r-1}} \cdot \frac{(r-1)!(13-r)!}{12!} \cdot \frac{4}{3}$ $= \frac{13-r}{r} \cdot \frac{4}{3}$ $\text{Coeff of } t_{r+1} \geq \text{Coeff of } t_r$ $4(13-r) \geq 3r$ $52 - 4r \geq 3r$ $7r \leq 52$ $r \leq 7$ $\therefore [\text{Coeff of } t_{r+1} \geq \text{Coeff of } t_r \text{ for } r \geq 8]$ $\text{Coeff of } t_8 > \text{Coeff of } t_7 > \dots > \text{Coeff of } t_1$ $\text{Also, Coeff of } t_{13} < \text{Coeff of } t_{12} < \dots < \text{Coeff of } t_8$ $T_8 = {}^{12}C_7 \cdot 3^5 \cdot (4x)^7$ $\therefore \text{Coeff. of } T_8 = {}^{12}C_7 \cdot 3^5 \cdot 4^7$ or greater coeff. = 3153199104	1 1 (2) 1 1	1 for $\frac{{}^{12}C_r}{{}^{12}C_{r-1}} \cdot \frac{4}{3}$ (2) (2)

Qn	Solutions	Marks	Comments: Criteria
13a.	 The only acceleration on the particle is gravity $\ddot{y} = -10$ Initially $t=0$ $\dot{x} = 20 \cos \alpha$ $\dot{y} = 20 \sin \alpha$ $\ddot{x} = 0$ $x = 0$ $y = 0$ $\ddot{y} = -10$ $\dot{y} = -10t + C$ $20 \sin \alpha = -10 \cdot 0 + C$ $\therefore \dot{y} = -10t + 20 \sin \alpha$ $y = -5t^2 + (20 \sin \alpha)t + C$ $t=0$ $y=0$ $\therefore C = 0$ $y = -5t^2 + 20t \sin \alpha \quad \text{--- (2)}$ eliminate t in (1) & (2) to get the Cartesian equation. $\textcircled{1} \quad t = \frac{x}{20 \cos \alpha}$ Sub in (2) $y = -5 \frac{x^2}{400 \cos^2 \alpha} + 20 \cdot \frac{x}{20 \cos \alpha} \cdot \sin \alpha$ $= -\frac{1}{80} \sec^2 \alpha x^2 + x \tan \alpha$ $= -\frac{1}{80} (1 + \tan^2 \alpha) x^2 + x \tan \alpha$ $= x \tan \alpha - \frac{x^2}{80} \sec^2 \alpha$	2	$-\frac{1}{2}$ PER ERROR $-\frac{1}{2}$ if didn't start from $\dot{x}=0, \dot{y}=-10$ $-\frac{1}{2}$ IF NO +C

Qn	Solutions	Marks	Comments: Criteria
	if $t = 100\alpha$ $y = x t - \frac{x^2}{80} (1+t^2)$ $10 = 20t - \frac{400(1+t^2)}{80}$ $10 = 20t - 5(1+t^2)$ $2 = 4t - 1 - t^2$ $t^2 - 4t + 3 = 0$ $(t-3)(t-1) = 0$ $t = 3$ $t = 1$ $\tan \alpha = 3$ $\tan \alpha = 1$ $\alpha = \tan^{-1} 3; 45^\circ$ $71^\circ 34'$	2	$\frac{1}{2}$ FOR CORRECT SUBSTITUTION $x=20$ $y=10$ $\frac{1}{2}$ FOR SIGNIFICANT SIMPLIFICATION
14	(b) $x = 3 \sin(2t + \alpha)$ $t=0$ $x = 1.5$ $1.5 = 3 \sin \alpha$ $\sin \alpha = \frac{1}{2}$ $\alpha = 30^\circ$	1	
(11)			
(12)	$\ddot{x} = 6 \cos(2t + \alpha)$ $\dot{x}' = -12 \sin(2t + \alpha)$ $= -4 (3 \sin(2t + \alpha))$ $= -4x$		

Qn	Solutions	Marks	Comments: Criteria
(iii)	$0 = 3 \sin(2t + \alpha)$ $\therefore 2t + \alpha = 0, \pi, 2\pi, \dots$ $2t + \frac{\pi}{6} = \pi$ $2t = \frac{5\pi}{6}$ $t = \frac{5\pi}{12}$	2	
(iv)	$-12 = -12 \sin(2t + \pi/6)$ $\sin(2t + \pi/6) = 1$ $2t + \pi/6 = \pi/2$ $t = \pi/6$	2	
(v)	$a = 7x$ $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 7x$ $\frac{1}{2} v^2 = \frac{7x^2}{2} + C$ $\frac{9}{2} = C$ $v^2 = 7x^2 + 9$ $v = \pm \sqrt{7x^2 + 9}$ $\therefore v = -\sqrt{7x^2 + 9}$	$t=0$ $x=0$ $v=3$ $x=0$ $v=3$	$v = \sqrt{7x}$ $1/3$ $u=3$
		3	

Qn	Solutions	Marks	Comments: Criteria
14c.	$f(x) = 1 - \cos \frac{x}{2}$ $T = \frac{2\pi}{1/2} = 4\pi$  $f^{-1}(x)$ exists $0 \leq x \leq 2\pi$ $(0 \text{ or } -2\pi \leq x \leq 0)$ $y = 1 - \cos \frac{x}{2}$ $x = 1 - \cos \frac{y}{2}$ $\cos \frac{y}{2} = 1 - x$ $\frac{y}{2} = \cos^{-1}(1 - x)$ $y = 2 \cos^{-1}(1 - x)$ 	① ① 1 $\frac{1}{2}$ $\frac{1}{2}$ ②	1 SHAPE 1 CORRECT DOMAIN/RANGE $-\frac{1}{2}$ no endpoints